11-6 求题 11-6 图所示电路在哪些频率时短路或开路。

(a) $Z = j\omega L - j\frac{1}{\omega C}$

短路：$Z=0 \Rightarrow \omega = \sqrt{\frac{1}{LC}}$

开路：$Z=\infty \Rightarrow \omega = \infty$

(b) $Y = j\omega C + \frac{1}{j\omega L} = 0$

开路 $Y=0 \Rightarrow \omega = \sqrt{\frac{1}{LC}}$

短路 $Y=\infty \Rightarrow \omega = 0$

(c) $Z = (j\omega L_1 - j\frac{1}{\omega C_1}) \parallel j\omega L_2$

$= \frac{j(\omega L_1 - \frac{1}{\omega C_1})\cdot \omega L_2}{\omega L_1 - \frac{1}{\omega C_1} + \omega L_2}$

短路：$Z=0 \Rightarrow \omega = \sqrt{\frac{1}{L_1 C}}$ or $\omega = 0$（分母为0）

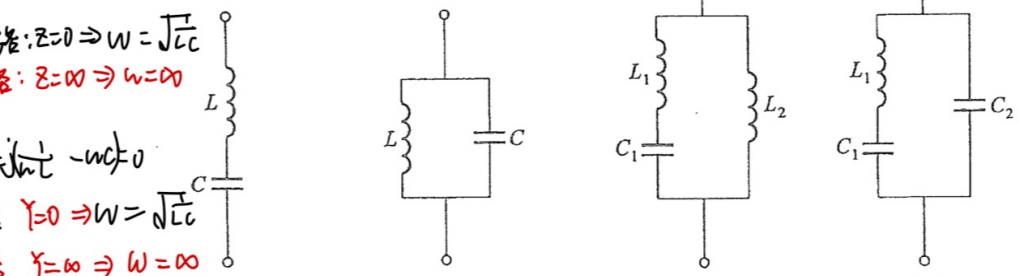
开路：$Z=\infty \Rightarrow \omega = \infty$ or $\omega = \sqrt{\frac{1}{(L_1+L_2)C}}$（分母为0）

(d) $Z = (j\omega L_1 - j\frac{1}{\omega C_1}) \parallel (j\frac{1}{\omega C_2})$

$= \frac{-(j\omega L_1 - j\frac{1}{\omega C_1})\cdot \frac{1}{\omega C_2}}{\omega L_1 - \frac{1}{\omega C_1} - \frac{1}{\omega C_2}} \approx \frac{1}{\omega}$

短路：$Z=0 \Rightarrow \omega = \sqrt{\frac{1}{L_1 C_1}}$ or $\omega = \infty$

开路：$Z=\infty \Rightarrow \omega = \sqrt{\frac{1}{C_1 L_1} + \frac{1}{C_2 L_1}}$ or $\omega = 0$



题 11-6 图

11-11 求题 11-11 图所示电路的谐振频率及各频段的电抗性质。

(a) $Z = \frac{1}{j\omega C_1} \parallel j\omega L_1 + j\omega L_2 \parallel \frac{1}{j\omega C_2}$

$= \frac{-\frac{L_1}{C_1}j}{(\omega L_1 - \frac{1}{\omega C_1})} - \frac{\frac{L_2}{C_2}j}{(\omega L_2 - \frac{1}{\omega C_2})}$

$= 0$

$\Rightarrow \omega_0 = \sqrt{\frac{L_1 - L_2}{L_1 L_2 (C_2 - C_1)}}$（整个电路串联谐振）

L_1与C_1并联谐振：$\omega_1 = \frac{1}{\sqrt{L_1 C_1}}$

L_2、C_2并联谐振：$\omega_2 = \frac{1}{\sqrt{L_2 C_2}}$

$0 < \omega < \omega_1$：感；$\omega_1 < \omega < \omega_2$：容；$\omega_2 < \omega < \infty$：感

$\omega_1 < \omega < \omega_0$：容；$\omega_0 < \omega < \omega_2$：感

开路 ω 在电感、电容各一个，短路相反？

(b) $Z = j\omega L + (\frac{1}{2}\cdot j\omega L) \parallel \frac{1}{j\omega C}$

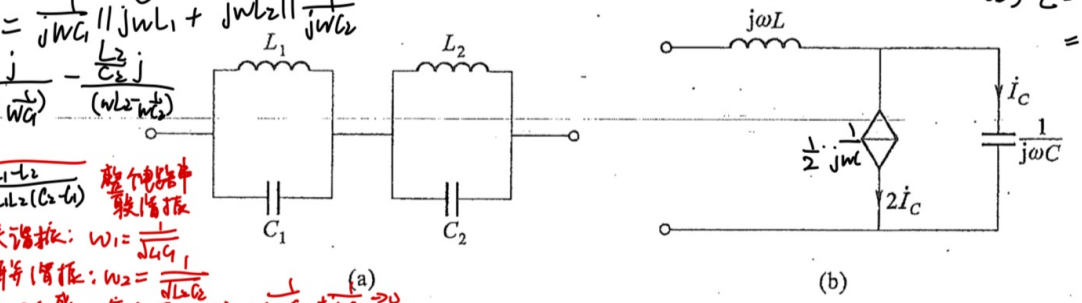
$= j\omega L + \frac{1}{3j\omega C} = 0$ 时

$\omega_0 = \frac{1}{\sqrt{3LC}}$

$\omega < \omega_0$：容性

$\omega > \omega_0$：感性

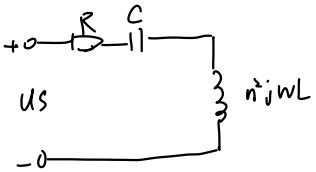
串联谐振，∴短路



题 11-11 图

11-13 题 11-13 图所示电路中 $R=10\ \Omega$，$C=0.1\ \mu F$，正弦电压 u_S 的有效值 $U_s = 1\ V$，电路的 Q 值为 100，求参数 L 和谐振时的 U_L。

等效电路：



$Z = R - j\frac{1}{\omega C} + n^2 j\omega L$

$-\frac{1}{\omega C} + n^2 \omega L = 0$ 时

$\omega = \frac{1}{n}\frac{1}{\sqrt{LC}}$

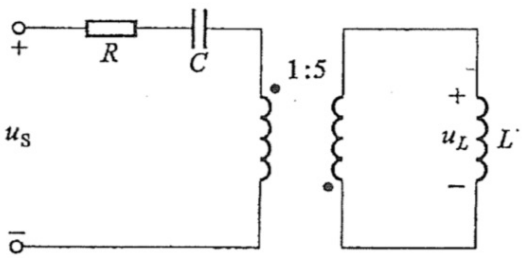
$Q = \frac{n^2 \omega L}{R} = 100$

$\Rightarrow L = 2.5H$ ✓

求的是L上的电压，非等效阻抗前

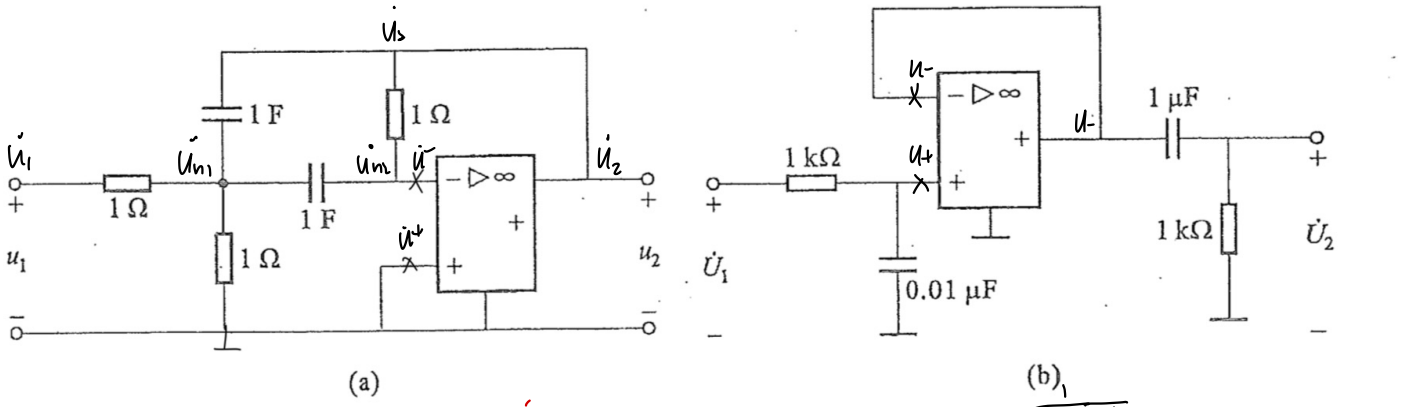
$U_L = nU_1 = 500\ V$

$U_L = 1\times 100 = 100V$ ✗



题 11-13 图

11-16 求题 11-16 图所示电路的转移电压比 $\frac{\dot{U}_2}{\dot{U}_1}$。



题 11-16 图

(a) $(1+1+\frac{1}{j\omega}+j\omega)\dot{U}_{n1}-\dot{U}_1-j\omega\dot{U}_2-j\omega\dot{U}_{n2}=0$

$(\frac{1}{j\omega}+1)\dot{U}_{n2}-j\omega\dot{U}_{n1}-\dot{U}_2=0$

$\dot{U}_{n2}=0$

$\Rightarrow \dot{U}_{n1}=-\dot{U}_2\cdot j\omega=\frac{\dot{U}_1+j\omega\dot{U}_2}{2+2j\omega}$

$\Rightarrow \frac{\dot{U}_2}{\dot{U}_1}=\frac{j\omega}{\omega^2-2-2j\omega}$

(b) $\dot{U}_+=\dot{U}_1\cdot\frac{\frac{1}{0.01j\omega\times10^6}}{1000+\frac{1}{0.01j\omega\times10^6}}$ $\dot{U}_+=\dot{U}_-$

$\dot{U}_2=\dot{U}_-\cdot\frac{1000}{\frac{1}{j\omega\times10^{-6}}+1000}$

$\Rightarrow \frac{\dot{U}_2}{\dot{U}_1}=\frac{1000+\frac{1}{0.01j\omega\times10^6}}{\frac{1}{0.01j\omega\times10^6}}\cdot\frac{\frac{1}{j\omega\times10^{-6}}+1000}{1000}$

$=1.01+j\omega\times10^{-5}-\frac{1000}{\omega}j$ $\frac{j\omega\times10^5}{(10^5+j\omega)(\omega^2+j\omega5)}$

13-7 已知一 RLC 串联电路的端口电压和电流为

$$u(t)=[100\cos(314t)+50\cos(942t-30°)]\ \text{V}$$

$$i(t)=[10\cos(314t)+1.755\cos(942t+\theta_3)]\ \text{A}$$

试求：(1) R、L、C 的值。

(2) θ_3 的值。

(3) 电路消耗的功率。

(1) $\omega=314$ rad/s 时

$R=\frac{U}{I}=\frac{100}{10}=10\Omega$

L与C发生谐振 $\Rightarrow \frac{1}{\omega C}=314$ rad/s

$\omega=942$ rad/s 时

$\dot{U}=25\sqrt{2}\angle-30°$ V

$\dot{I}=1.241\angle\theta_3$ A

$\frac{\dot{U}}{\dot{I}}=R+j\omega L+\frac{1}{j\omega C}$

$28.49\angle-30°-\theta_3$

$=10+j942L+\frac{1}{j942C}$

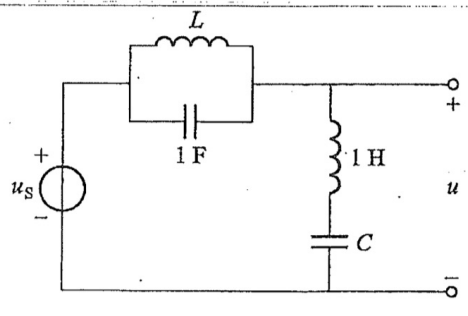
$\Rightarrow L=0.0319$ H ✓

$C=3.18\times10^{-4}$ F

(2) $P=\frac{100}{\sqrt{2}}\times\frac{10}{\sqrt{2}}+\frac{50}{\sqrt{2}}\times\frac{1.755}{\sqrt{2}}\times\cos(-30°-\theta_3)$

$=715.4$ W ✓

13-9 题 13-9 图所示电路中 $u_s(t)$ 为非正弦周期电压，其中含有 $3\omega_1$ 及 $7\omega_1$ 的谐波分量。如果要求在输出电压 $u(t)$ 中不含这两个谐波分量，那么 L、C 应为多少？



题 13-9 图

$3\omega_1$ 或 $7\omega_1$ 时 LC 发生谐振，$\dot{U}=0$ 或 L与C发生谐振，$\dot{U}=0$

$\Rightarrow j\omega L=j\frac{1}{\omega C}$

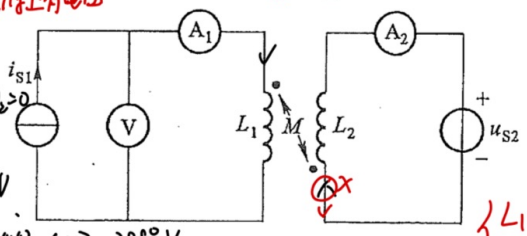
$\omega=\frac{1}{\sqrt{LC}}=\frac{1}{\sqrt{LC}}$ 和 $\frac{1}{\sqrt{LC}}$

$\frac{1}{\sqrt{LC}}=3\omega_1$ $L=\frac{1}{9\omega_1^2}$

$\frac{1}{\sqrt{LC}}=7\omega_1$ $C=\frac{1}{49\omega_1^2}$ ✓

2° $C=\frac{1}{9\omega_1^2}$ $L=\frac{1}{49\omega_1^2}$

13-10 题 13-10 图所示电路中 $i_{S1}=[5+10\cos(10t-30°)-5\sin(30t+60°)]$ A，$u_{S2}(t)=300\sin(10t)+150\cos(30t-30°)$ V，$L_1=L_2=2\text{H}$，$M=0.5\text{H}$。求图中交流电表的读数和电源发出的功率 P。



题 13-10 图

$\dot{I}_{A1}=\sqrt{5^2+(\frac{5}{2})^2+(\frac{5\sqrt{3}}{2})^2}=9.35\text{A}$

~~$j\omega L_1\cdot\dot{I}_{S1}\oplus j\omega M\cdot\dot{I}_2=0$~~ 电流源上有电压

$\dot{U}_{S2}\ominus j\omega L_2\cdot\dot{I}_2+j\omega M\cdot\dot{I}_1=0$

$i_{S1}=5\text{A}$，$u_{S2}=0$ 时 $V=0$，$A_1=5\text{A}$，$A_2=0$

$\omega=1\text{rad/s}$ 时

$\dot{I}_{S1}=5\sqrt{2}\angle-30°\text{A}$，$\dot{U}_{S2}=150\sqrt{2}\angle-90°\text{V}$

$\dot{I}_2=9.119\angle-174.437°\text{A}$ $\dot{U}=104.443\angle47.388°\text{V}$

$\omega=3\text{rad/s}$ 时

$\dot{I}_{S1}=2.5\sqrt{2}\angle-30°\text{A}$ $\dot{U}_{S1}=75\sqrt{2}\angle-30°\text{V}$

$\dot{I}_2=\frac{5\sqrt{14}}{8}\angle-110.538°\text{A}$

$\dot{U}=79.672\angle40.585°\text{V}$

$\Rightarrow A_1$ 读数：9.354A ✓ A_2：10.586A ✗

V：131.362V ✗

$$\begin{cases}L_1\frac{di_{S1}}{dt}-M\frac{di_2}{dt}=u_1\\-M\frac{di_{S1}}{dt}+L_2\frac{di_2}{dt}=u_{S2}\end{cases}$$

$\Rightarrow i_2=\frac{1}{L_1}\int u_{S2}dt+\frac{M}{L_2}i_{S1}\Rightarrow u_1=\cdots\Rightarrow A_2:9.424$ $V:261.86\text{V}$

$P_{iS1}=-93.8\text{W}$

$P_{uS2}=93.8\text{W}$